

Module 9 & 10 Summary (Probability Distributions & Statistical Inference)

MODULE 9: PROBABILITY DISTRIBUTIONS

Key Idea: A probability distribution shows how values of a random variable are distributed.

Types:

1. Discrete: Countable values (e.g., 0,1,2...)
2. Continuous: Any value in a range

Important Distributions:

Bernoulli: One trial (success/failure). Mean = p , Variance = $p(1-p)$

Example: Toss a coin once.

Binomial: Repeated trials. Mean = np , Variance = $np(1-p)$

Example: 10 coin tosses, count heads.

Poisson: Events over time. Mean = λ

Example: Calls per minute.

Normal Distribution: Bell-shaped curve. Uses mean (μ) and standard deviation (σ).

68% rule: within 1σ , 95% within 2σ .

Z-score: $z = (x - \mu) / \sigma$

Exponential: Time between events. Mean = $1/\lambda$

How to Identify:

- Fixed trials → Binomial
- Time events → Poisson
- Measurements → Normal

MODULE 10: STATISTICAL INFERENCE

Key Idea: Use sample data to make conclusions about a population.

Point Estimation: Single value estimate (e.g., mean)

Example: Sample mean = $\text{sum}(x)/n$

Good Estimator Properties:

- Unbiased
- Consistent
- Efficient

Interval Estimation:

Confidence Interval = estimate $\pm$ margin of error

Example:

Mean CI = $\bar{x} \pm z(\sigma/\sqrt{n})$

Hypothesis Testing:

Steps:

1. State H_0 and H_1
2. Choose α
3. Compute test statistic
4. Decision

Errors:

Type I: Reject true H_0

Type II: Accept false H_0

Tests:

Z-test: large sample / known σ

T-test: small sample

Chi-square: variance/categorical

P-value:

If $p \leq \alpha \rightarrow$ reject H_0

If $p > \alpha \rightarrow$ fail to reject H_0

Example:

If $p = 0.03$ and $\alpha = 0.05 \rightarrow$ reject H_0